OrBit User Guide

Module 05: P2P Repository Seeding & Bootstrapping

PUBLIC DOCUMENTATION — FOR DEVELOPERS USERS

1 Introduction to P2P Bootstrapping

When a developer joins an existing OrBit project team or checks out a workspace for the first time, they need the full repository history and baseline files. In traditional systems, this requires running a heavy `git clone` from a centralized server (e.g., GitHub).

In OrBit, repository initialization uses **P2P Bootstrapping**: the joining developer downloads an encrypted, signed Git packfile directly from an online team member (a **Seeder Peer**) over the local network or P2P mesh network.

2 The Seeding Bootstrap Lifecycle

The P2P bootstrap procedure follows a rigorous multi-stage verification pipeline managed by `request_bootstrap` in the Rust daemon:

Stage	State Name	Operation & Action Executed
1	`Downloading`	Client sends `BootstrapNetworkMessage::TransferRequest` to seeder peer over `libp2p`.
2	`Verifying`	Seeder packages `shadow.git` objects into a packfile, encrypts it with `project_token`, signs it with Ed25519 key, and streams it back. Client verifies signature.
3	`Restoring`	Client imports packfile atomically into local `.orbit/shadow.git` using `ShadowRepo::import_packfile_atomic`.
4	`Completed`	Workspace files are checked out to disk, snapshot manifests are indexed, and project is active.

Table 1: OrBit Bootstrap Status Pipeline

3 Step-by-Step Guide: Joining a Project via P2P

3.1 1. Accepting a Team Invite

1. Your team admin invites you to a project via OrBit Desktop Client. 2. The project metadata and shared `project_token` are synced to your client via the Control Server.

3.2 2. Triggering P2P Bootstrap

1. On the Projects view, click **Join & Bootstrap Repository**. 2. Select your desired local workspace folder. 3. The client calls Tauri command `request_bootstrap({ project_id })`. 4. OrBit locates active online seeders using libp2p mDNS local discovery or Kademlia DHT.

> The downloading peer verifies the seeder's Ed25519 cryptographic signature against the downloaded packfile bytes before importing. If signature validation fails or tampered bytes are detected, bootstrap aborts immediately to prevent bad code injection.

3.3 3. Automatic Workspace Initialization

Once verification succeeds:

- `.orbit/shadow.git` is populated with team history up to generation N.
- Both `main` and `sub` snapshot manifests are generated.
- The file watcher is attached to your local workspace folder.